

Muhammad Haroon Khan

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EDUCATION

Ph.D. Physics	Aug 2022 – Present	University of Houston, Houston, TX
M.S. Physics	Aug 2022 – Aug 2024	University of Houston, Houston, TX
M. Phil Physics	Feb 2020 – Apr 2022	Quaid-I-Azam University, Islamabad
B.S. Physics	Sep 2015 – Jan 2020	Quaid-I-Azam University, Islamabad

HONORS & AWARDS

❖ Graduate Research Assistantship	2022 – 2027	Higher Education Commission of Pakistan, USA
❖ Research Assistantship	2024 – Present	University of Houston, Houston, TX, USA
❖ TcSUH Graduate Student Travel Support Award	2026 – 2027	Texas Center for Superconductivity, University of Houston
❖ TEX-E Fellowship	2026 – 2027	TEX-E, USA
❖ Best Presentation Award, Experimental Physics Session	2025	University of Houston Physics Research Day, USA
❖ Departmental Fellowship for Academic Excellence	2022	Quaid-i-Azam University, Islamabad

WORK EXPERIENCE

Graduate Student Researcher	Aug 2023 – Present	University of Houston, TX
❖ Supervisor: Dr. Shuo Chen ❖ Investigating high-performance Mg_3Sb_2-based thermoelectric materials for low-grade industrial waste heat recovery through transition-metal doping, defect engineering, phase composition optimization and conversion efficiency device fabrication. ❖ Investigating interface defects and microstructural analysis of Hg1223-based high-temperature superconducting 2024single crystals using advanced TEM techniques, including diffraction and energy dispersive X-ray spectroscopy. ❖ Developing novel, low-cost electrocatalysts for green hydrogen production through electrochemical water-splitting, correlating structural properties via XRD and TEM with catalytic performance and stability in industrial-scale water electrolyzer systems.		
Teaching Assistant	Fall 2024	University of Houston, TX
❖ Teaching Assistant in College Physics Lab 1 ❖ Demonstrated to students their weekly lab experiments, conducted weekly office hours, class lab report grading, and exam grading.		
Master Student Researcher	Feb 2021 – Apr 2022	Quaid-I-Azam University

- ❖ Supervisor: Dr. Hassnain Jaffri
- ❖ Investigated Ba-doped and Ba–Ti co-doped $BiFeO_3$ thin films synthesized via the chemical solution deposition method. Performed Structural and microstructural characterizations using XRD, Raman spectroscopy, SEM and studied optical, electrical and magnetic properties.

RESEARCH PUBLICATIONS

Published

- ❖ **Muhammad Haroon Khan**, Rui Shu, Shuo Chen, “Phase Segregation in High Mn-doped Mg_3SbBi Based Thermoelectrics”. *ACS Applied Energy Materials* (2025)
- ❖ Mumtaz, Fiza, **Muhammad Haroon Khan**, and G. Hassnain Jaffari. "Correlation between the composition, phase, band structure, ferroelectric and leakage responses of $Bi_{1-x}Ba_xFe_{1-y}Ti_yO_3$ thin films." *Thin Solid Films* (2022).
- ❖ Publications can be accessed on my [Google Scholar Profile](#).

Manuscripts in Preparation

- ❖ **Muhammad Haroon Khan**, L. Deng, S. Chen et al., “Pressure Induced Edge Dislocations Study of Pristine and Pressure Quenched Hg-1223 Superconducting Materials.” Manuscript in preparation.
- ❖ P. Byaruhanga, **Muhammad Haroon Khan**, S. Chen et al., “Synergistic Sulfur-Selenium Co-Doping of NiFe LDH Enhanced Oxygen Evolution.” Manuscript in preparation

PRESENTATIONS AND WORKSHOPS

- ❖ **Muhammad Haroon Khan**, Shuo Chen, “ Modulating phase purity boosts thermoelectric performance in high Mn-doped $Mg_3(Sb, Bi)_2$ -based materials ” The 42nd International Conference on Thermoelectrics 2026.
- ❖ **Muhammad Haroon Khan**, Rui Shu, Shuo Chen “ Phase segregation in high Mn-doped Mg_3SbBi thermoelectric materials”. *American Physical Society & TSAAPT Joint Spring Meeting* (2025).
- ❖ Paul Byaruhanga, **Muhammad Haroon Khan**, Shuo Chen “Investigating Effect of Dual Doping S and Se on NiFe Layered Double Hydroxide Catalyst for Oxygen Evolution Reaction”. *Electrochemical Society Meeting* (2025).
- ❖ Tech boot camp on robophysics Summer 2024 **Georgia Institute of Technology, GA, USA**
- ❖ IPOLS Scientific Programming Summer 2024 **Emory University, GA, USA**
in Python Bootcamp
- ❖ International School on Rietveld Refinement & Paid Distribution ([ISBARM-22](#)) Summer 2022 **National Centre for Physics (NCP), PK**

RELAVENT SKILLS

Material Synthesis

- ❖ Nanostructured thermoelectric materials synthesis and processing.
- ❖ Device fabrication and thermoelectric module assembly.
- ❖ Catalyst synthesis for alkaline water electrolysis.
- ❖ Thin-film fabrication via sol-gel deposition and spin coating.
- ❖ Annealing Furnaces
- ❖ Vacuum Deposition Systems
- ❖ Spin Coater

Electron Microscopy & Sample Preparation

- ❖ Mechanical polishing and cross-sectional TEM sample preparation.

- ❖ Focused Ion Beam (FIB) milling.
- ❖ Precision ion polishing using PIPS II
- ❖ Dimpling and thinning techniques.

Material Characterizations

- ❖ Scanning Electron Microscopy (SEM)
- ❖ Transmission Electron Microscopy (TEM)
- ❖ Raman Spectroscopy
- ❖ Thermoelectric Property Measurements
- ❖ Thermoelectric Device Performance Testing
- ❖ Magnetic Measurements
- ❖ Ultraviolet–Visible (UV–Vis) Spectroscopy
- ❖ X-Ray Diffraction (XRD)

Scientific Data Analysis & Programming

- ❖ ImageJ, FullProf / Rietveld Refinement
- ❖ Computational material database analysis using Materials Project
- ❖ DigitalMicrograph (TEM)
- ❖ Python
- ❖ MATLAB
- ❖ LabVIEW
- ❖ Arduino

ADDITIONAL COURSES & CERTIFICATES

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| ❖ Introduction to Programming with MATLAB
<i>Vanderbilt University</i> | Aug. 2020 |
| ❖ Semiconductor Physics and Devices
<i>UC Boulder</i> | Oct. 2020 |
| ❖ How Things Work: An Introduction to Physics
<i>University of Virginia</i> | July. 2020 |
| ❖ Fundamentals of Optics by ANSYS | July 2025 |

LEADERSHIP AND VOLUNTEERING EXPERIENCE

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| ❖ TEX-E Fellow | 2026 - 2027 | TEX-E
UH Energy Coalition |
| ❖ Vice Chair of Mentorship Committee | Fall 2025 - Present | |
| ❖ Assistant Session Chair | May 17–21, 2026 | International Thermoelectric Conference 2026, Houston
Department of Physics at the University of Houston
UH Energy Coalition |
| ❖ NSM Grad. Student Council Representative | 2024 - 2025 | |
| ❖ Vice Chair of Sustainability Committee | Spring 2025 | SEFH, Houston |
| ❖ Judge in Physics and Astronomy category at Science & Engineering Fair Houston | 2022 - 2026 | |